

ISRO Computer Science Mock Test 3 - 2025

Instructions:

- **Total Questions:** 95 (80 Technical + 15 Aptitude)
- **Time:** 120 minutes
- **Marking:** +1 for correct, -1/3 for wrong, 0 for unattempted
- **All questions are compulsory**

PART A - TECHNICAL QUESTIONS (80 Questions)

Advanced Data Structures and Graph Theory (Q1-18)

****1.**** In a Fibonacci heap, what is the amortized time complexity of decrease-key operation?
(a) $O(1)$ (b) $O(\log n)$ (c) $O(n)$ (d) $O(\sqrt{n})$

****2.**** Which data structure supports range minimum queries in $O(1)$ time after $O(n \log n)$ preprocessing?
(a) Segment tree (b) Sparse table (c) Fenwick tree (d) Cartesian tree

****3.**** In a trie with n strings of maximum length m , what is the space complexity?
(a) $O(nm)$ (b) $O(n + m)$ (c) $O(\text{ALPHABET_SIZE} \times nm)$ (d) $O(n^2m)$

****4.**** The planarity of a graph can be tested in:
(a) $O(n^2)$ (b) $O(n \log n)$ (c) $O(n)$ (d) $O(n^3)$

****5.**** In a balanced binary search tree, the time complexity for range queries is:
(a) $O(\log n + k)$ (b) $O(n)$ (c) $O(k \log n)$ (d) $O(\log^2 n)$

****6.**** Which graph algorithm can detect negative cycles?
(a) Dijkstra's (b) Floyd-Warshall (c) Prim's (d) Kruskal's

****7.**** The maximum matching in a bipartite graph can be found in:
(a) $O(VE)$ (b) $O(V^2E)$ (c) $O(V^3)$ (d) $O(E^2)$

****8.**** In a skip list, the expected search time is:
(a) $O(1)$ (b) $O(\log n)$ (c) $O(\sqrt{n})$ (d) $O(n)$

****9.**** The vertex connectivity of a complete graph K_n is:
(a) $n-1$ (b) n (c) $n-2$ (d) $\lfloor n/2 \rfloor$

****10.**** Which data structure is optimal for implementing a priority queue for Dijkstra's algorithm?
(a) Binary heap (b) Fibonacci heap (c) Array (d) Linked list

****11.**** In a 2-3 tree, what is the minimum number of keys in a non-root node?
(a) 1 (b) 2 (c) 3 (d) 0

****12.**** The time complexity of Tarjan's algorithm for finding articulation points is:

- (a) $O(V + E)$ (b) $O(V^2)$ (c) $O(VE)$ (d) $O(E \log V)$

****13.**** Which property is maintained in a treap?

- (a) BST property only (b) Heap property only (c) Both BST and heap properties (d) Neither

****14.**** The maximum number of edges in a simple planar graph with n vertices is:

- (a) $3n - 6$ (b) $2n - 4$ (c) $n - 1$ (d) $n(n-1)/2$

****15.**** In union-find with both path compression and union by rank, the time complexity is:

- (a) $O(\log n)$ (b) $O(\alpha(n))$ (c) $O(1)$ (d) $O(\sqrt{n})$

****16.**** The girth of a graph is the length of its:

- (a) Longest path (b) Shortest cycle (c) Diameter (d) Spanning tree

****17.**** Which data structure supports both predecessor and successor queries efficiently?

- (a) Hash table (b) van Emde Boas tree (c) Bloom filter (d) Count-min sketch

****18.**** In a suffix tree, the space complexity for a string of length n is:

- (a) $O(n)$ (b) $O(n^2)$ (c) $O(n \log n)$ (d) $O(1)$

Advanced Algorithms and Complexity (Q19-30)

****19.**** The traveling salesman problem is:

- (a) In P (b) NP-Complete (c) NP-Hard but not NP-Complete (d) PSPACE-Complete

****20.**** Which approximation algorithm has a 2-approximation ratio for the metric TSP?

- (a) Nearest neighbor (b) Minimum spanning tree (c) Christofides algorithm (d) Greedy algorithm

****21.**** The maximum clique problem can be solved in:

- (a) Polynomial time (b) Exponential time (c) Logarithmic time (d) Constant time

****22.**** In randomized algorithms, Las Vegas algorithms guarantee:

- (a) Correctness (b) Running time (c) Both (d) Neither

****23.**** The complexity class EXPTIME contains problems solvable in:

- (a) Polynomial time (b) Exponential time (c) Logarithmic space (d) Constant space

****24.**** Which problem is PSPACE-Complete?

- (a) SAT (b) Hamiltonian path (c) Quantified Boolean Formula (d) Vertex cover

****25.**** The approximation ratio of the greedy algorithm for set cover is:

- (a) 2 (b) $\ln n$ (c) $\sqrt{n}$ (d) n

****26.**** In parallel algorithms, what does NC stand for?

(a) Non-deterministic Complete (b) Nick's Class (c) Polynomial time with polylog processors
(d) Not Computable

****27.**** The edit distance between two strings of length m and n can be computed in:

(a) $O(mn)$ (b) $O(\min(m,n))$ (c) $O(\max(m,n))$ (d) $O(m + n)$

****28.**** Which technique is used in Karger's minimum cut algorithm?

(a) Dynamic programming (b) Random contraction (c) Greedy approach (d) Divide and conquer

****29.**** The vertex cover problem admits a:

(a) PTAS (b) FPTAS (c) 2-approximation (d) No constant approximation

****30.**** In streaming algorithms, what is the primary constraint?

(a) Time (b) Space (c) Processor count (d) Network bandwidth

Machine Learning and Artificial Intelligence (Q31-40)

****31.**** In neural networks, what is the purpose of the sigmoid activation function?

(a) Introduce non-linearity (b) Normalize inputs (c) Prevent overfitting (d) Reduce complexity

****32.**** Which learning algorithm is guaranteed to find the global optimum?

(a) Gradient descent (b) K-means (c) Linear regression (convex) (d) Neural networks

****33.**** In decision trees, what does information gain measure?

(a) Tree depth (b) Reduction in entropy (c) Number of leaves (d) Accuracy

****34.**** The curse of dimensionality primarily affects:

(a) Linear algorithms (b) Distance-based algorithms (c) Tree-based algorithms (d) Rule-based algorithms

****35.**** In reinforcement learning, what does the Q-function represent?

(a) Policy function (b) Value of state-action pair (c) Reward function (d) Transition probability

****36.**** Which kernel is commonly used in Support Vector Machines?

(a) Linear (b) RBF (c) Polynomial (d) All of the above

****37.**** In unsupervised learning, what is the goal of PCA?

(a) Classification (b) Dimensionality reduction (c) Clustering (d) Regression

****38.**** Overfitting in machine learning can be reduced by:

(a) Regularization (b) Cross-validation (c) Dropout (d) All of the above

****39.**** In Bayesian networks, nodes represent:

(a) Dependencies (b) Random variables (c) Conditional probabilities (d) Evidence

****40.**** The perceptron learning algorithm is guaranteed to converge if data is:
(a) Linearly separable (b) Normally distributed (c) High-dimensional (d) Binary

Software Engineering and System Design (Q41-50)

****41.**** In microservices architecture, what is the primary challenge?
(a) Development complexity (b) Service communication (c) Data consistency (d) All of the above

****42.**** Which design pattern is used to create objects without specifying their exact class?
(a) Singleton (b) Factory (c) Observer (d) Strategy

****43.**** In agile methodology, what is the purpose of a "sprint retrospective"?
(a) Plan next sprint (b) Review completed work (c) Improve process (d) Estimate effort

****44.**** What is the main principle of Domain-Driven Design (DDD)?
(a) Code organization (b) Business domain focus (c) Technical architecture (d) Performance optimization

****45.**** In continuous integration/deployment, what is a "pipeline"?
(a) Data flow (b) Automated workflow (c) Network connection (d) Code repository

****46.**** Which architectural pattern separates concerns into layers?
(a) MVC (b) Microservices (c) Event-driven (d) Layered architecture

****47.**** In software testing, what is "mutation testing"?
(a) Testing different inputs (b) Testing code variants (c) Performance testing (d) Security testing

****48.**** What is the primary goal of DevOps?
(a) Cost reduction (b) Development-Operations collaboration (c) Code quality (d) Security enhancement

****49.**** In distributed systems, what is "eventual consistency"?
(a) Immediate consistency (b) Consistency achieved over time (c) No consistency (d) Perfect consistency

****50.**** Which metric measures code complexity?
(a) Lines of code (b) Cyclomatic complexity (c) Number of functions (d) Execution time

Database Systems and Big Data (Q51-60)

****51.**** In ACID properties, what does "Isolation" ensure?
(a) Transaction completion (b) Concurrent transaction independence (c) Data persistence (d) Constraint satisfaction

****52.**** Which join algorithm has the best worst-case performance?
(a) Nested loop join (b) Hash join (c) Sort-merge join (d) Index nested loop join

****53.**** In data warehousing, what is a "star schema"?

(a) Database design pattern (b) Query optimization (c) Indexing strategy (d) Replication method

****54.**** What is the main advantage of column-store databases?

(a) OLTP performance (b) Analytical query performance (c) Storage efficiency (d) Concurrent access

****55.**** In MapReduce, what does the "shuffle" phase do?

(a) Data input (b) Mapper output sorting (c) Reducer execution (d) Final output

****56.**** Which NoSQL database type is best for hierarchical data?

(a) Key-value (b) Document (c) Column-family (d) Graph

****57.**** In database indexing, what is a "covering index"?

(a) Index on all columns (b) Index containing all query columns (c) Primary index (d) Clustered index

****58.**** What is "sharding" in distributed databases?

(a) Data replication (b) Horizontal partitioning (c) Vertical partitioning (d) Query optimization

****59.**** In OLAP cubes, what is "drilling down"?

(a) Data aggregation (b) Moving to detailed data (c) Data filtering (d) Schema modification

****60.**** Which property is NOT guaranteed by BASE (in NoSQL)?

(a) Basically available (b) Soft state (c) Eventual consistency (d) ACID compliance

Computer Networks and Distributed Systems (Q61-70)

****61.**** In network protocols, what is the purpose of a "keep-alive" message?

(a) Data transmission (b) Connection maintenance (c) Error detection (d) Flow control

****62.**** Which consensus algorithm is used in blockchain?

(a) Paxos (b) Raft (c) Proof of Work (d) Byzantine fault tolerance

****63.**** In network security, what is "perfect forward secrecy"?

(a) Immediate key destruction (b) Session key independence (c) Unlimited key lifetime (d) Key backup security

****64.**** What is the main purpose of Content Delivery Networks (CDN)?

(a) Security (b) Reduced latency (c) Data backup (d) Load balancing

****65.**** In distributed computing, what is the "two generals problem"?

(a) Resource allocation (b) Consensus with unreliable communication (c) Load balancing (d) Fault detection

****66.**** Which protocol ensures reliable multicast communication?

(a) UDP (b) TCP (c) Reliable UDP (d) IGMP

****67.**** In peer-to-peer networks, what is "churn"?

(a) Data corruption (b) Node joining/leaving (c) Network congestion (d) Protocol overhead

****68.**** What is the purpose of network address translation (NAT)?

(a) Speed enhancement (b) IP address conservation (c) Security improvement (d) Protocol conversion

****69.**** In quality of service (QoS), what does "jitter" refer to?

(a) Packet loss (b) Delay variation (c) Bandwidth utilization (d) Error rate

****70.**** Which routing algorithm adapts to network changes?

(a) Static routing (b) Dynamic routing (c) Source routing (d) Default routing

Operating Systems and Computer Architecture (Q71-80)

****71.**** In computer architecture, what is "branch prediction"?

(a) Jump instruction optimization (b) Predicting conditional branch outcomes (c) Loop optimization (d) Function call optimization

****72.**** Which memory consistency model is strongest?

(a) Sequential consistency (b) Weak consistency (c) Release consistency (d) Eventual consistency

****73.**** In virtual memory, what is "copy-on-write"?

(a) Immediate page copying (b) Delayed page copying (c) Read-only pages (d) Write-through caching

****74.**** What is the purpose of cache coherence protocols?

(a) Cache performance (b) Multi-processor cache consistency (c) Cache size optimization (d) Cache replacement

****75.**** In operating systems, what is a "system call"?

(a) Function call (b) Kernel service request (c) Interrupt handler (d) Library function

****76.**** Which scheduling algorithm is optimal for minimizing average turnaround time?

(a) FCFS (b) Shortest job first (c) Round robin (d) Priority scheduling

****77.**** In memory management, what causes "internal fragmentation"?

(a) Variable-sized allocation (b) Fixed-sized allocation waste (c) External memory pressure (d) Cache misses

****78.**** What is "speculative execution" in processors?

(a) Parallel execution (b) Executing before certainty (c) Error correction (d) Power optimization

****79.**** In file systems, what is "journaling"?

(a) Access logging (b) Transaction logging (c) Performance monitoring (d) Security auditing

****80.**** Which technique improves processor performance through parallelism?

(a) Pipelining (b) Superscalar execution (c) Out-of-order execution (d) All of the above

PART B - APTITUDE/ABILITY TEST (15 Questions)

****81.**** In a certain code, FLOWER is written as EKNVDQ. How is GARDEN written?

(a) FZQCDK (b) FZQCDM (c) GZQCDM (d) FZRCDK

****82.**** What is the next number in the sequence: 2, 6, 12, 20, 30, ?

(a) 40 (b) 42 (c) 44 (d) 48

****83.**** A train 100m long crosses a platform 200m long in 15 seconds. What is the speed of the train?

(a) 72 km/h (b) 60 km/h (c) 54 km/h (d) 48 km/h

****84.**** If $\log_3(x) = 2$, then x equals:

(a) 6 (b) 8 (c) 9 (d) 12

****85.**** In a group of 40 students, 24 play cricket, 20 play football, and 8 play both. How many play neither?

(a) 4 (b) 6 (c) 8 (d) 12

****86.**** What is the value of: $2^3 + 3^2 - 5^1$?

(a) 12 (b) 14 (c) 16 (d) 18

****87.**** If the ratio of ages of A and B is 3:4 and A is 15 years old, what is B's age?

(a) 18 (b) 20 (c) 22 (d) 24

****88.**** Complete the pattern: JAK, KBL, LCM, MDN, ?

(a) NEO (b) NEP (c) OEP (d) NFO

****89.**** What is 30% of 150?

(a) 40 (b) 45 (c) 50 (d) 55

****90.**** A cube has a volume of 64 cm^3 . What is the length of its edge?

(a) 4 cm (b) 6 cm (c) 8 cm (d) 16 cm

****91.**** If $3x - 7 = 14$, then x equals:

(a) 5 (b) 6 (c) 7 (d) 8

****92.**** What day of the week will it be 45 days from Tuesday?

(a) Monday (b) Tuesday (c) Friday (d) Saturday

****93.**** The sum of interior angles of a hexagon is:

(a) 540° (b) 720° (c) 900° (d) 1080°

****94.**** If a number increased by 25% becomes 75, what is the original number?

(a) 50 (b) 55 (c) 60 (d) 65

****95.**** In the series 3, 6, 11, 18, 27, ?, what comes next?

(a) 36 (b) 38 (c) 40 (d) 42

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